Ex. No: 15

Title: BPTT Flow and Gradient Analysis

Packages: TensorFlow, Keras, NumPy, Matplotlib

AIM:

To analyse Backpropagation Through Time (BPTT) in an RNN by finding the gradient at every time step.

ALGORITHM:

1. Start the program and import the required libraries.
2. Create a random input sequence of 20 time steps and a target value.
3. Create a SimpleRNN layer and a Dense layer.
4. Use tf.GradientTape to record the forward pass and calculate the loss.
5. Find the gradient of the loss with respect to the input at every time step (BPTT).
6. Print the gradient value of each time step.
7. Plot the gradient magnitude against the time step and stop.

PROGRAM:

```
import numpy as np
import matplotlib.pyplot as plt
import tensorflow as tf
from tensorflow import keras
from tensorflow.keras import layers

tf.random.set_seed(0)
steps = 20
x = tf.random.normal((1, steps, 1))
target = tf.constant([[1.0]])

rnn = layers.SimpleRNN(8, activation="tanh")
dense = layers.Dense(1)

# Backpropagation Through Time (BPTT): gradient of loss w.r.t. each input
with tf.GradientTape() as tape:
    tape.watch(x)
    output = dense(rnn(x))
    loss = tf.reduce_mean((output - target) ** 2)

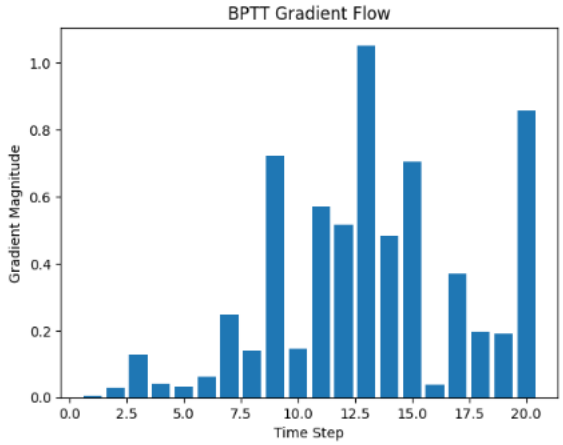
grads = tape.gradient(loss, x)
norms = np.abs(grads.numpy()[0, :, 0])

for t in range(steps):
    print("Time step", t + 1, "- Gradient =", float(norms[t]))

plt.bar(range(1, steps + 1), norms)
plt.xlabel("Time Step")
plt.ylabel("Gradient Magnitude")
plt.title("BPTT Gradient Flow")
plt.show()
```

OUTPUT:

```
Time step 1 - Gradient = 0.006178565323352814
Time step 2 - Gradient = 0.02850986458361149
Time step 3 - Gradient = 0.12679821252822876
Time step 4 - Gradient = 0.04065901041030884
Time step 5 - Gradient = 0.031569622457027435
Time step 6 - Gradient = 0.061471059918403625
Time step 7 - Gradient = 0.24604517221450806
Time step 8 - Gradient = 0.13837654888629913
Time step 9 - Gradient = 0.7224694490432739
Time step 10 - Gradient = 0.14622554183006287
Time step 11 - Gradient = 0.569257915019989
Time step 12 - Gradient = 0.5170969367027283
Time step 13 - Gradient = 1.052039384841919
Time step 14 - Gradient = 0.4827105402946472
Time step 15 - Gradient = 0.7035592198371887
Time step 16 - Gradient = 0.03874558210372925
Time step 17 - Gradient = 0.36834990978240967
Time step 18 - Gradient = 0.19590970873832703
Time step 19 - Gradient = 0.19154492020606995
Time step 20 - Gradient = 0.8570795059204102
```



RESULT:

BPTT was analysed successfully. The gradients of the earlier time steps are much smaller than those of the later time steps, which shows the vanishing gradient problem in RNNs.